

ERAM RAHMAN

US Citizen | San Diego, CA (UC San Diego) | Los Angeles, CA | Open to Relocation |
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EDUCATION

Expected Graduation: 2027

Bachelor of Science in Computer Engineering — University of California San Diego

Major: Computer Engineering

Relevant Coursework: Data structures and Algorithms, Software Engineering, Digital Systems, Computer Architecture, Operating Systems

SKILLS

Programming Languages:

- Python, C++, C, Assembly (ARM), HTML, CSS, Java, JavaScript, MATLAB

Web and Application Development:

- Front-End Development (React, HTML, CSS), Next.js (Full-Stack, API Routes), REST API Integration, Git+GitHub, Figma, SQL, JavaFX, GitHub Project, GitHub Actions CI, JUnit

Other Skills:

- Custom PC Assembly & System Integration, Hardware Diagnostics & Troubleshooting, Technical Consulting, Small Business Ownership & Operations, Altium Design, Agile Software Processing, Behavior-Driven Development (BDD)

EXPERIENCE

Undergraduate Researcher — Robot Perception & Mapping

- Conduct research on robot perception and mapping in unknown environments, focusing on simultaneous localization and mapping (SLAM), semantic mapping, and uncertainty-aware decision making.
- Study methods that integrate visual, depth, inertial, and semantic sensor data to build consistent 3D representations of dynamic environments.

PC Technician and Owner

- Provided technical consulting and support to 100+ clients, diagnosing hardware and software issues
- Built and troubleshoot custom PCs for gaming, school, and work
- Provided technical support and hardware recommendations

Tri Meadows Healthcare, IT Volunteer

- Provided on-site and remote technical support for staff and community members, troubleshooting hardware, software, and network issues\
- Assisted users with operating systems, application installs, account access, and basic cybersecurity best practices

PROJECTS

PC Recommendation Website

- Developed a full-stack web application using Next.js (React) with server-side API routes for personalized PC recommendations
- Implemented a questionnaire-based system to match users with optimal builds based on budget and use case
- Integrated external APIs to dynamically fetch product data, pricing, and specifications

Voice Controlled Navigation

- Built a web-based navigation application using Mapbox API with real-time GPS tracking and routing
- Implemented voice command functionality to enable hands-free navigation and improve user safety
- Developed dynamic route recalculation and multi-stop navigation features

Student Clock Application

- Developed a Java-based productivity application using JavaFX and SQLite for task and time management
- Implemented priority-based scheduling to optimize study time allocation
- Designed an MVC architecture with persistent data storage and real-time tracking features